

Ogan K. Kumova, PhD

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PROFESSIONAL SUMMARY

Virologist and immunologist with over a decade of hands-on research experience spanning neonatal immunology, infectious disease, and vaccine development. Expertise in murine in vivo models, multiparameter flow cytometry (BD Fortessa certified), viral propagation, and innate and adaptive immune assays. Postdoctoral training at the FDA developing and validating immunological endpoints for novel vaccine platforms within a NIH-funded multi-institutional cooperative research center. Author of 15 peer-reviewed articles and Contributing Editor on 2 National Academies reports. Experienced collaborator and mentor in multi-site research environments.

RESEARCH EXPERTISE

In Vivo & Infectious Disease: Influenza, Legionella, Neisseria gonorrhoeae · Murine neonatal/ascending infection models · Viral propagation (TCID50, plaque, neutralization) · IACUC protocol development

Immunology & Assay Development: Multicolor flow cytometry (BD Fortessa Certified) · Intracellular cytokine staining · ELISA/ELISpot/multiplex cytokine profiling · T- and B-cell phenotyping · Mucosal and systemic immunity

Vaccine Science & Genomics: Liposome/OMV vaccine candidates · Adjuvant evaluation · Antigen formulation · Route-of-immunization studies · RNA-seq · NanoString · Single-cell sequencing · RT-qPCR · ddPCR · Western blot · Cloning

Data & Operations: FlowJo · GraphPad Prism · Python · R · LIMS/Quartz · GLP/SOP development · PMP coursework completed

PROFESSIONAL EXPERIENCE

Program Officer, Vaccine Safety

February 2023 – Present

National Academies of Sciences, Engineering, and Medicine — Washington, D.C.

- Manage expert committee consensus studies on vaccine safety and immunology for federal agency sponsors (VA, DoD, CDC, FDA), overseeing the full study lifecycle: expert recruitment, committee coordination, meeting logistics, evidence synthesis, and report production.
- Currently supporting two active ad hoc studies: (1) Neurodegenerative Outcomes and Selected Military Exposures; and (2) Examination of the Direct Neurocognitive Effects of SARS-CoV-2 Infection Across the Lifespan and the Neuroprotective Effects of COVID Vaccination.
- Served as Contributing Editor on two published National Academies reports: Evidence Review of the Adverse Effects of COVID-19 Vaccination and Intramuscular Vaccine Administration (National Academies Press, 2024) and Vaccine Risk Monitoring and Evaluation at the Centers for Disease Control and Prevention (National Academies Press, 2025).
- Delivered invited briefing on COVID-19 vaccine safety to Congressional staff of the U.S. House and Senate Appropriations Committees and the Senate HELP Committee (October 2025).
- Manage project timelines, vendor contracts, and budget tracking across concurrent study portfolios; coordinates multi-stakeholder processes across federal sponsors, committee members, external reviewers, and institutional staff.

ORISE Postdoctoral Research Fellow

October 2020 – February 2023

U.S. Food and Drug Administration, Office of Vaccines Research and Review — Silver Spring, MD · Supervisor:

Margaret C. Bash, MD, MPH · NIH-U19 Gonorrhea Vaccine Cooperative Research Center

- Led development and characterization of liposome- and outer membrane vesicle (OMV)-based Neisseria gonorrhoeae vaccine candidates, including antigen selection, formulation optimization, and adjuvant evaluation.
- Designed and executed multiparameter flow cytometry panels and immunology assays to assess systemic and mucosal B- and T-cell responses, antibody subclasses, and innate immune activation after immunization.

- Conducted murine ascending infection immunization studies, including colony maintenance, IACUC protocol management, inoculations, reproductive tract and lymphoid tissue harvests, blood collection, and downstream endpoint analysis.
- Evaluated route-of-immunization effects on anti-gonococcal immune responses; results published in *Journal of Infectious Diseases* (2025; PMID 40971931).
- Collaborated across NIH-U19 multi-site consortium; contributed to grant progress reports, data sharing, and consortium meetings.
- Gained direct exposure to the FDA regulatory environment, including GLP standards, SOP development, regulatory documentation, and OVR's framework for preclinical vaccine safety and immunogenicity evaluation.

Doctoral Researcher

August 2015 – July 2020

Drexel University College of Medicine — Philadelphia, PA · PI: Alison Carey, MD · Dissertation: Altered Innate Immune Responses to Influenza in the Neonate

- Identified type I IFN overactivation and oxidative stress as primary drivers of immunopathology in neonatal influenza infection (*Mucosal Immunology* 2022; PMID 36352099).
- Demonstrated that intranasal *Lactobacillus rhamnosus* GG restores transcriptional responsiveness and early viral control in neonatal lungs; performed RNA-seq and pathway analysis (*PLoS Pathogens* 2019; PMID 31603951).
- Designed and executed multiparameter flow cytometry panels to characterize neonatal T-cell activation, effector function, CD31 expression, and cytokine production in murine lung and spleen.
- Characterized neonatal CD8⁺ T-cell defects, TCR repertoire dynamics in extremely preterm neonates, and role of CD31 in effector CTL trafficking; contributed to five peer-reviewed publications.
- Performed ELISA, ELISpot, cytokine multiplex, immunofluorescence, and qRT-PCR; managed murine breeding colonies and neonatal handling, intranasal inoculation, BAL collection, and tissue harvest.

Laboratory Manager

December 2014 – August 2015

Drexel University College of Medicine — Philadelphia, PA · Supervisor: Alison Carey, MD

- Managed all daily operations of an active neonatal immunology research laboratory, including regulatory compliance (IACUC, IBC), equipment maintenance and calibration, and supply management.
- Oversaw laboratory budget: tracked expenditures against NIH grant accounts, reconciled credit card transactions, processed purchase orders, and liaised with departmental finance and university procurement.
- Planned and executed a full laboratory relocation, coordinating the safe transport of equipment, reagents, biological materials, and records while maintaining research continuity.

Laboratory Technician II

December 2012 – December 2014

Drexel University College of Medicine — Philadelphia, PA · Supervisor: Shira Ninio, PhD

- Investigated *Legionella pneumophila* pathogenesis — biofilm formation, OmpA-like protein characterization, and intracellular replication — contributing to peer-reviewed publications (*FEMS Microbiology Letters* 2016; *Biomolecules* 2022).
- Received and managed an independent grant to develop a murine co-infection model of influenza and *L. pneumophila*, including experimental design, protocol development, and model optimization.

EDUCATION

Doctor of Philosophy, Microbiology & Immunology

2015 – 2020

Drexel University College of Medicine — Philadelphia, PA

GPA: 4.0 · Minor: Health Administration

Master of Science, Infectious Diseases

2013 – 2015

Drexel University College of Medicine — Philadelphia, PA

GPA: 3.97

PEER-REVIEWED PUBLICATIONS

15 peer-reviewed articles. * denotes co-first authorship.

- Rao AS, Onufer AP, **Kumova O**, Roberts LTN, Mossabeh R, Pascasio J, Carey AJ. Myeloid TLR2 signaling amplifies immunopathology in influenza-infected murine neonates. *Am J Respir Cell Mol Biol*. 2026;aanag050. PMID: 42089286.
- Matthias KA, Connolly KL, Jerse AE, **Kumova OK**, Macintyre AN, Gray MC, Thomas KS, Criss AK, Zielke RA, Li L, Sikora AE, Ramirez-Bencomo F, Thistlethwaite A, Derrick JP, Lu WE, Bash MC. Evaluation of immunization route in induction of vaccine-mediated anti-gonococcal immune responses in a murine model of ascending infection. *J Infect Dis*. 2025;232(5):e765–e777. PMID: 40971931.
- Rao AS, Ugwu N, Onufer AP, **Kumova O**, Carey AJ. Cathelicidin-related antimicrobial peptide (CRAMP) is toxic during neonatal murine influenza virus infection. *J Immunol*. 2025;214(5):1022–1031. PMID: 40101750.
- Kumova OK**, Galani IE, Rao A, Johnson H, Triantafyllia V, Matt SM, Pascasio J, Gaskill PJ, Andreaskos E, Katsikis PD, Carey AJ. Severity of neonatal influenza infection is driven by type I interferon and oxidative stress. *Mucosal Immunol*. 2022;15(6):1309–1320. PMID: 36352099.
- Marin C, **Kumova OK**, Ninio S. Characterization of a novel regulator of biofilm formation in the pathogen *Legionella pneumophila*. *Biomolecules*. 2022;12(2):225. PMID: 35204726.
- Das P, Acharya S, Prahaladan VM, **Kumova OK**, Malaeb S, Behera S, Agarwal B, Christensen DJ, Carey AJ, Bhandari V. Chitin-derived AVR-48 prevents experimental bronchopulmonary dysplasia and BPD-associated pulmonary hypertension in newborn mice. *Int J Mol Sci*. 2021;22(16):8547. PMID: 34445253.
- Chaubey S, Nader YM, Shah D, **Kumova OK**, Prahaladan VM, Carey AJ, Andersson S, Bhandari V. α 1,3-Fucosyltransferase-IX, an enzyme of pulmonary endogenous lung stem cell marker SSEA-1, alleviates experimental bronchopulmonary dysplasia. *Pediatr Res*. 2021;89(5):1126–1135. PMID: 32303051.
- Gary E, O'Connor M, Chakhtoura M, Tardif V, **Kumova OK**, Malherbe DC, Sutton WF, Haigwood NL, Kutzler MA, Haddad EK. Adenosine deaminase-1 enhances germinal center formation and functional antibody responses to HIV-1 envelope DNA and protein vaccines. *Vaccine*. 2020;38(22):3821–3831. PMID: 32280045.
- Kumova OK**, Fike AJ, Thayer JL, Nguyen LT, Mell JC, Pascasio J, Stairiker C, Leon LG, Katsikis PD, Carey AJ. Lung transcriptional unresponsiveness and loss of early influenza virus control in infected neonates is prevented by intranasal *Lactobacillus rhamnosus* GG. *PLoS Pathog*. 2019;15(10):e1008072. PMID: 31603951.
- Kumova OK***, Fike AJ*, Carey AJ. Dissecting the defects in the neonatal CD8⁺ T-cell response. *J Leukoc Biol*. 2019;106(5):1051–1061. PMID: 31260598. (Review)
- Fike AJ, **Kumova OK**, Tardif VJ, Carey AJ. Neonatal influenza-specific effector CTLs retain elevated CD31 levels at the site of infection and have decreased IFN- γ production. *J Leukoc Biol*. 2019;105(3):539–549. PMID: 30536476.
- Carey AJ, Hope JL, Mueller YM, Fike AJ, **Kumova OK**, van Zessen DBH, Steegers EAP, van der Burg M, Katsikis PD. Public clonotypes and convergent recombination characterize the naïve CD8⁺ T-cell receptor repertoire of extremely preterm neonates. *Front Immunol*. 2017;8:1859. PMID: 29312340.
- Fike AJ, Nguyen LT, **Kumova OK**, Carey AJ. Characterization of CD31 expression on murine and human neonatal T lymphocytes during development and activation. *Pediatr Res*. 2017;82(1):133–140. PMID: 28355204.
- Goodwin IP, **Kumova OK**, Ninio S. A conserved OmpA-like protein in *Legionella pneumophila* required for efficient intracellular replication. *FEMS Microbiol Lett*. 2016;363(16):fnw173. PMID: 27421957.
- Carey AJ, Gracias DT, Thayer JL, Boesteanu AC, **Kumova OK**, Mueller YM, Hope JL, Fraietta JA, van Zessen DB, Katsikis PD. Rapid evolution of the CD8⁺ TCR repertoire in neonatal mice. *J Immunol*. 2016;196(6):2602–2613. PMID: 26873987.

National Academies Reports — Contributing Editor

Ancona D, **Kumova OK**, Stratton K, Henney JE, editors. Vaccine Risk Monitoring and Evaluation at the Centers for Disease Control and Prevention. Washington, DC: National Academies Press; 2025. doi:10.17226/29240. PMID: 41289413.

Rosenberg D, **Kumova OK**, Stratton K, Bass AR, editors. Evidence Review of the Adverse Effects of COVID-19 Vaccination and Intramuscular Vaccine Administration. Washington, DC: National Academies Press; 2024. doi:10.17226/27746. PMID: 39312602.

HONORS & FUNDING

- VesiVax Vaccine Formulation Against Neisseria gonorrhoeae (SBIR)
- Oak Ridge Institute for Science and Education (ORISE) Fellowship — \$98,000/year, 3 years
- Drexel University Bondi Fellowship for Outstanding Graduate Student, 2019 — \$2,500
- Drexel University Dean's Fellowship for Excellence in Collaborative Research, 2017–2018 — \$65,000
- PAESM Infection and Immunity 2016 Poster Award
- NIH Clinical and Translational Research Workshop 2016 Cohort

CERTIFICATIONS & CREDENTIALS

BD Fortessa Certified User & Multi-Parameter Panel Designer · Category B Infectious Agent Shipping · CRLA Level II Certified Tutor · GLP Training & SOP Development · PMP Coursework Completed (Exam Pending) · First Aid & CPR · Bilingual: Turkish and English